

# Paper 1 - Set 12 - Solutions

Generated practice paper: answer key and worked solutions.

## Answer key

| Question | Answer | Topic                                    |
|----------|--------|------------------------------------------|
| 1        | E      | Trigonometric functions and equations    |
| 2        | B      | Integration                              |
| 3        | D      | Ratio, proportion and percentages        |
| 4        | F      | Circles and circle theorems              |
| 5        | C      | Functions and transformations            |
| 6        | D      | Number, units and estimation             |
| 7        | B      | Exponentials and logarithms              |
| 8        | F      | Indices and surds                        |
| 9        | B      | Trigonometric functions and equations    |
| 10       | A      | Graphs of functions                      |
| 11       | B      | Quadratic functions                      |
| 12       | A      | Angles, polygons, congruence and vectors |
| 13       | A      | Simultaneous equations and inequalities  |
| 14       | B      | Sequences and series                     |
| 15       | C      | Circles and circle theorems              |
| 16       | A      | GCSE algebra                             |
| 17       | A      | Differentiation                          |
| 18       | F      | Quadratic functions                      |
| 19       | A      | Integration                              |
| 20       | C      | Binomial expansion                       |

## Worked solutions

### Question 1 TRG-11 · Trigonometric functions and equations · spec MM4.4

The function  $f(x) = 3\sin(2x) - 1$  is defined for all real  $x$ . Find the difference between its maximum and minimum values.

**Answer: E**

Range is  $[-4, 2]$ ; the difference is 6.

### Question 2 INT-06 · Integration · spec MM7.5

The trapezium rule with 2 strips of equal width is used to estimate  $\int_0^2 x^2 dx$ . Which of the following is true?

**Answer: B**

$h = 1$ :  $\frac{h}{2}[y_0 + 2y_1 + y_2] = \frac{1}{2}[0 + 2(1) + 4] = 3$ . The true value is  $\frac{8}{3}$ ; the curve bends upwards (convex), so the chords lie above it and the rule overestimates.

### Question 3 RAT-01 · Ratio, proportion and percentages · spec M3.8

The price of a coat is reduced by 20% in a sale, and then the sale price is reduced by a further 25%. The final price is what percentage of the original price?

**Answer: D**

$0.8 \times 0.75 = 0.6$ , so 60%.

### Question 4 CIR-08 · Circles and circle theorems · spec MM3.3

A chord of a circle of radius 13 is at perpendicular distance 5 from the centre. Find the length of the chord.

**Answer: F**

The perpendicular from the centre bisects the chord: half-length  $= \sqrt{13^2 - 5^2} = 12$ , so the chord is 24.

### Question 5 FUN-03 · Functions and transformations · spec MM8.2

Let  $f(x) = 2x + 1$  and  $g(x) = x^2 - 3$ . Find  $f(g(2)) - g(f(2))$ .

**Answer: C**

$g(2) = 1$ ,  $f(1) = 3$ .  $f(2) = 5$ ,  $g(5) = 22$ . So  $3 - 22 = -19$ .

### Question 6 NUM-07 · Number, units and estimation · spec M1.2

A car travels at 90 km/h. What is this speed in metres per second?

**Answer: D**

90 000 m in 3600 s:  $90000/3600 = 25$  m/s.

### Question 7 LOG-04 · Exponentials and logarithms · spec MM5.2

Given that  $\log_a x = 3$  and  $\log_a y = 5$ , find  $\log_a \left( \frac{x^2 \sqrt{y}}{a} \right)$ .

**Answer: B**

$2\log_a x + \frac{1}{2}\log_a y - \log_a a = 6 + \frac{5}{2} - 1 = \frac{15}{2}$ .

**Question 8** IND-21 · Indices and surds · spec MM1.1

If  $x = 2^p$ ,  $y = 2^q$  and  $\frac{x^3}{y^2} = 2^5$  with  $xy = 2^{10}$ , find  $p$ .

**Answer: F**

$3p - 2q = 5$  and  $p + q = 10$ . From the second,  $q = 10 - p$ ; then  $3p - 20 + 2p = 5$ , so  $5p = 25$  and  $p = 5$ .

**Question 9** TRG-09 · Trigonometric functions and equations · spec MM4.5

Simplify  $\frac{1 - \cos^2 \theta}{\cos \theta \tan \theta}$  for values of  $\theta$  where the expression is defined.

**Answer: B**

$\cos \theta \tan \theta = \sin \theta$ , and  $1 - \cos^2 \theta = \sin^2 \theta$ . So the expression is  $\sin \theta$ .

**Question 10** GRA-03 · Graphs of functions · spec MM8.1

Which of the following could be the equation of a cubic curve that touches the  $x$ -axis at  $x = 2$  and crosses it at  $x = -1$ , and has a negative  $y$ -intercept?

**Answer: A**

Touching at 2 needs the factor  $(x - 2)^2$  and crossing at  $-1$  the factor  $(x + 1)$ .  $y = (x - 2)^2(x + 1)$  has  $y$ -intercept  $4 > 0$ , so the sign is reversed:  $y = -(x - 2)^2(x + 1)$  with intercept  $-4$ .

**Question 11** QUA-11 · Quadratic functions · spec MM1.3

The curve  $y = x^2 - 2ax + a^2 + a - 6$  has its vertex on the  $x$ -axis. Find the value of  $a$ .

**Answer: B**

Complete the square:  $y = (x - a)^2 + a - 6$ . The vertex is  $(a, a - 6)$ , which lies on the  $x$ -axis when  $a = 6$ .

**Question 12** GEO-14 · Angles, polygons, congruence and vectors · spec M5.5

In triangle  $ABC$ ,  $D$  lies on  $AB$  and  $E$  lies on  $AC$  with  $DE$  parallel to  $BC$ .  $AD = 4$ ,  $DB = 6$  and  $DE = 5$ . Find  $BC$ .

**Answer: A**

Triangles  $ADE$  and  $ABC$  are similar with ratio  $AD : AB = 4 : 10$ .  $BC = 5 \times \frac{10}{4} = 12.5$ .

**Question 13** SIM-03 · Simultaneous equations and inequalities · spec MM1.5

Find the set of values of  $x$  satisfying both  $3x - 7 < 2$  and  $x^2 > 4$ .

**Answer: A**

$3x < 9 \Rightarrow x < 3$ .  $x^2 > 4 \Rightarrow x < -2$  or  $x > 2$ . Combining:  $x < -2$  or  $2 < x < 3$ .

**Question 14** SEQ-07 · Sequences and series · spec MM2.1

A sequence satisfies  $x_{n+1} = \frac{1}{1 - x_n}$  with  $x_1 = 2$ . Find  $x_{100}$ .

**Answer: B**

$x_2 = \frac{1}{1-2} = -1$ ,  $x_3 = \frac{1}{1-(-1)} = \frac{1}{2}$ ,  $x_4 = \frac{1}{1-\frac{1}{2}} = 2 = x_1$ : the sequence has period 3. Since  $100 = 3 \times 33 + 1$ ,  $x_{100} = x_1 = 2$ .

**Question 15** CIR-16 · Circles and circle theorems · spec MM3.2

The point  $P(x, y)$  moves so that its distance from  $(3, 0)$  is twice its distance from  $(0, 0)$ . The locus of  $P$  is a circle. Find its radius.

**Answer: C**

$$(x-3)^2 + y^2 = 4(x^2 + y^2) \Rightarrow 3x^2 + 3y^2 + 6x - 9 = 0 \Rightarrow x^2 + 2x + y^2 = 3 \Rightarrow (x+1)^2 + y^2 = 4. \text{ Radius } 2.$$

**Question 16** ALG-03 · GCSE algebra · spec M4.6

Simplify  $\frac{x^2 - 9}{x^2 + x - 12} \times \frac{x^2 - 16}{x^2 - 3x}$ .

**Answer: A**

Factorise everything:  $\frac{(x-3)(x+3)}{(x+4)(x-4)} \times \frac{(x-4)(x+4)}{x(x-3)}$ . The  $(x+4)$  factors cancel and one  $(x-3)$  cancels, leaving  $\frac{(x+3)}{x(x-4)}$ .

**Question 17** DIF-08 · Differentiation · spec MM6.2

Given  $f(x) = x^{\frac{3}{2}} - 6x^{\frac{1}{2}}$  for  $x > 0$ , find the value of  $x$  at which  $f$  has its minimum.

**Answer: A**

$$f'(x) = \frac{3}{2}x^{1/2} - 3x^{-1/2} = \frac{3}{2\sqrt{x}}(x-2) = 0 \text{ at } x=2; f' \text{ changes from negative to positive there, so it is a minimum.}$$

**Question 18** QUA-20 · Quadratic functions · spec MM1.3

For how many integer values of  $k$  does  $x^2 + kx + 12 = 0$  have integer roots?

**Answer: F**

Integer roots  $r, s$  with  $rs = 12$ : pairs  $(1, 12), (2, 6), (3, 4)$  and their negatives, giving  $k = -(r+s) = -13, -8, -7, 13, 8, 7$ : six values.

**Question 19** INT-14 · Integration · spec MM7.5

The trapezium rule is used to estimate  $\int_a^b f(x) dx$ . For which of the following functions is the estimate guaranteed to be an underestimate for every choice of  $a < b$ ?

**Answer: A**

The trapezium rule underestimates when the curve is concave (bends downwards) so that chords lie below the curve:  $y = -x^2$  everywhere.  $x^3$  changes concavity at 0.

**Question 20** BIN-15 · Binomial expansion · spec MM2.4

The coefficient of  $x^2$  in  $(1+x)^n + (1+x)^{n+1}$  is 36. Find  $n$ .

**Answer: C**

$$\binom{n}{2} + \binom{n+1}{2} = \frac{n(n-1) + (n+1)n}{2} = n^2 = 36, \text{ so } n = 6.$$